DOE ASPECT · DE-FOA-0003646 · TEAMING PARTNER LIST TPL-0000073

Bibliography and References Cited

Not requested by the teaming form. Held with the package so the technical claims in the profile can be traced by anyone who asks for them.

4	1	1
PEER-REVIEWED SOURCES	WITH DR. MA AS AUTHOR	PROVISIONAL PATENT

ENZYMATIC DEPOLYMERIZATION OF PET

1 Lu, H.; Diaz, D. J.; Czarnecki, N. J.; Zhu, C.; Kim, W.; Shroff, R.; Acosta, D. J.; Alexander, B. R.; Cole, H. O.; Zhang, Y.; Lynd, N. A.; Ellington, A. D.; Alper, H. S. **Machine learning-aided engineering of hydrolases for PET depolymerization.** *Nature* 2022, 604 (7907), 662-667.
DOI 10.1038/s41586-022-04599-z · PMID 35478237

> Source of FAST-PETase. PlastiBioFuel did not develop this enzyme and makes no claim to it. Cited to fix the boundary between prior art and the company contribution.

2 Tournier, V.; Topham, C. M.; Gilles, A.; David, B.; Folgoas, C.; Moya-Leclair, E.; Kamionka, E.; Desrousseaux, M.-L.; Texier, H.; Gavalda, S.; Cot, M.; Guemard, E.; Dalibey, M.; Nomme, J.; Cioci, G.; Barbe, S.; Chateau, M.; Andre, I.; Duquesne, S.; Marty, A. **An engineered PET depolymerase to break down and recycle plastic bottles.** *Nature* 2020, 580 (7802), 216-219.
DOI 10.1038/s41586-020-2149-4 · PMID 32269349

> Benchmark for enzymatic PET depolymerization productivity, and the competing route that returns terephthalate to virgin-equivalent PET rather than to fuel.

3 Yoshida, S.; Hiraga, K.; Takehana, T.; Taniguchi, I.; Yamaji, H.; Maeda, Y.; Toyohara, K.; Miyamoto, K.; Kimura, Y.; Oda, K. **A bacterium that degrades and assimilates poly(ethylene terephthalate).** *Science* 2016, 351 (6278), 1196-1199.
DOI 10.1126/science.aad6359 · PMID 26965627

> Origin of the PETase and MHETase system, and the source of the terephthalic acid and ethylene glycol product pair that the company intends to route onward.

ENZYME IMMOBILIZATION IN FRAMEWORK MATERIALS

4 Wang, X.; Lan, P. C.; **Ma, S.** **Metal-Organic Frameworks for Enzyme Immobilization: Beyond Host Matrix Materials.** *ACS Central Science* 2020, 6 (9), 1497-1506.
DOI 10.1021/acscentsci.0c00687 · PMID 32999925

> Authored from the Department of Chemistry, University of North Texas. Dr. Shengqian Ma is the PlastiBioFuel subaward lead for framework and immobilization chemistry under Sponsored Research Agreement SRA1016.

PLASTIBIOFUEL INTELLECTUAL PROPERTY

5 Ward, M. D. **U.S. Provisional Patent Application 63/848,456**, filed July 2025.

> Covers framework immobilization, spiral-flow reactor geometry, and the integrated depolymerization process.

VERIFICATION

> References 1 through 4 were retrieved from PubMed and verified by identifier on August 25, 2026.

PlastiBioFuel LLC · michael@plastibiofuel.com · 817-319-7383 UEI KQAWZ54RDUM8 · CAGE 175D6